Insurance Risk Prediction Models

Evaluation Report: Regression, Classification and Explainability

Course	Data Science Advanced Programming (IBM)
Assignment	Assignment 3 - Insurance Risk Prediction Models
Dataset	med-insurance.csv (1,338 policyholders; 7 columns)
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Executive summary

This report builds and compares machine-learning models that predict how much a policyholder will cost an insurer (regression) and whether the policyholder is high risk (classification). The best pricing model, **XGBoost**, explains **92.2%** of the variation in expenses with a test RMSE of **$3,364** and MAE of **$2,120**, against an RMSE of $5,086 for plain linear regression. For risk triage, a **Random Forest with class weights** identifies 24 of the 27 high-risk customers in the test set (recall 88.9%, precision 92.3%, ROC-AUC 0.993). SHAP analysis shows that **smoking status, age and BMI** drive almost all of the predictions, which gives the insurer a transparent, auditable explanation for every price.

Deliverables accompanying this report: **01_Regression_Insurance_Expenses.ipynb** and **02_Classification_Insurance_Risk.ipynb** (fully executed, with outputs).

1. Business Problem and Data

1.1 Why risk prediction matters

An insurer collects **premiums** from many customers and pays **claims** to the few who suffer adverse events (risk pooling). **Underwriting** is the process of judging how risky a customer is in order to decide whether to insure them and at what price. If prices are too low for high-risk customers, the insurer loses money; if they are too high for safe customers, those customers leave. Two prediction tasks support this decision:

- **Regression:** predict the continuous yearly cost (*expenses*) of a customer so premiums can be set fairly.
- **Classification:** flag customers who are likely to be high risk, so safe customers can be approved automatically and risky ones sent to a human underwriter.

1.2 Dataset

The dataset contains 1,337 policyholders (one exact duplicate row was removed; no missing values were found) with six predictors: *age*, *sex*, *bmi*, *children*, *smoker* and *region*. The target is *expenses*, the yearly medical cost charged to the insurer. Mean expenses are $13,279 but the median is only $9,386, and the skewness is 1.52: a few very expensive customers pull the average up. About 20% of customers are smokers, and they cost on average **$32,050** versus **$8,441** for non-smokers (3.8 times more).

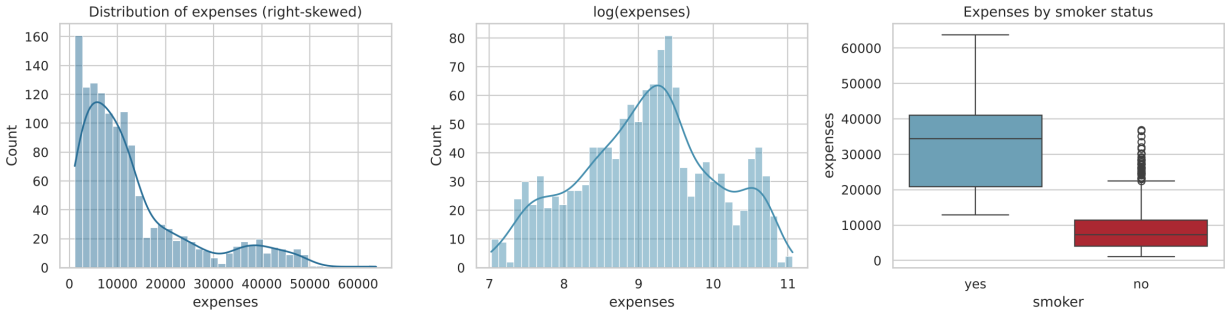


Figure 1. Expenses are right-skewed (left, centre); smokers have far higher costs (right).



Figure 2. Cost rises with age; the highest costs are obese (BMI >= 30) smokers. Region and sex show little effect.

1.3 Key EDA findings

- **Smoker** is by far the strongest driver of cost; **age** is second and rises steadily.
- **BMI** matters mainly for obese smokers - an *interaction* that a plain linear model cannot represent.
- **Sex, region and number of children** have weak relationships with cost.

- Large claims (above the IQR upper fence) were kept because they are genuine insurance events, not data errors.

2. Methodology

2.1 Preprocessing and validation design

- **Split:** 80 % train (1069 rows) / 20 % test (268 rows), stratified (by smoker for regression, by risk label for classification). The test set is used only once, for final reporting.
- **Encoding:** sex, smoker and region are one-hot encoded (first level dropped). Numeric variables are standardised for the linear models (their penalties depend on coefficient size); tree models use unscaled features, which keeps SHAP plots readable.
- **No leakage:** encoders, scalers and resamplers sit inside sklearn/imblearn *Pipelines*, so they are fitted on training folds only.
- **Cross-validation:** 5-fold CV (KFold for regression, StratifiedKFold for classification) with GridSearchCV for hyperparameter tuning on training data only.

2.2 Regression models

Model	Idea	Tuned hyperparameters (best found)
Linear Regression	Baseline; coefficients = average change in cost per unit change, others held constant	none
Ridge (L2)	Shrinks coefficients; copes with correlated features	alpha = 0.1
Lasso (L1)	Shrinks and can zero out coefficients (feature selection)	alpha = 1
Elastic Net	Mix of L1 and L2	alpha = 0.01, l1_ratio = 0.8
Random Forest	Many decorrelated trees on bootstrap samples, predictions averaged	max_depth=4, max_features=1.0, min_samples_leaf=5, n_estimators=400
XGBoost	Trees built sequentially, each correcting the previous errors	learning_rate=0.02, max_depth=3, n_estimators=200, subsample=0.8

2.3 Classification set-up

The dataset has no ready-made risk label, so **High Risk** was defined as yearly expenses in the **top 10 %** (at or above $34,833); this is an explicit modelling assumption. It produces a realistically imbalanced problem: 10.0% of customers are high risk. *Expenses* was removed from the inputs to avoid target leakage. A model that always predicts 'low risk' would score 89.9% accuracy while catching **no** high-risk customer, so accuracy alone is not a meaningful metric.

Four classifiers (Logistic Regression, Decision Tree, Random Forest, XGBoost) were each trained under four imbalance strategies: **no handling** (baseline), **class weights** (XGBoost: scale_pos_weight = 9.0), **SMOTE** oversampling and **random undersampling**. Resampling was applied only inside the training folds.

2.4 Evaluation metrics

- **MAE** - average dollar error; easiest for business readers.
- **RMSE** - squares errors first, so one badly mis-priced large claim is heavily penalised; the metric closest to an insurer's solvency concern.

- **R-squared / Adjusted R-squared** - share of cost variation explained; the adjusted version penalises useless variables.
- **Precision** - of customers flagged high risk, how many truly are (low precision = annoying safe customers with extra reviews).
- **Recall** - of truly high-risk customers, how many were caught (low recall = under-priced risky customers and real losses). Recall is prioritised, so models were selected on the cross-validated **F2 score**.
- **ROC-AUC** - ranking quality across all possible thresholds.

3. Regression Results: Predicting Insurance Expenses

Table 1 reports performance on the 268 unseen test customers. CV RMSE is the 5-fold cross-validated RMSE on the training data and shows stability.

Model	MAE ($)	RMSE ($)	R2	Adj. R2	CV RMSE ($)
Linear Regression	3,592	5,086	0.821	0.815	6,297 +/- 629
Ridge (L2)	3,593	5,086	0.821	0.815	6,297 +/- 629
Lasso (L1)	3,592	5,086	0.821	0.815	6,297 +/- 629
Elastic Net	3,614	5,092	0.820	0.815	6,298 +/- 635
Random Forest	2,178	3,400	0.920	0.917	4,749 +/- 612
XGBoost	**2,120**	**3,364**	**0.922**	**0.919**	**4,722 +/- 648**
Linear + smoker/obesity interaction*	2,140	3,440	0.918	0.915	-

Table 1. Regression model comparison (test set). *Extra experiment: linear regression with two engineered features (obese flag and smoker x obese).

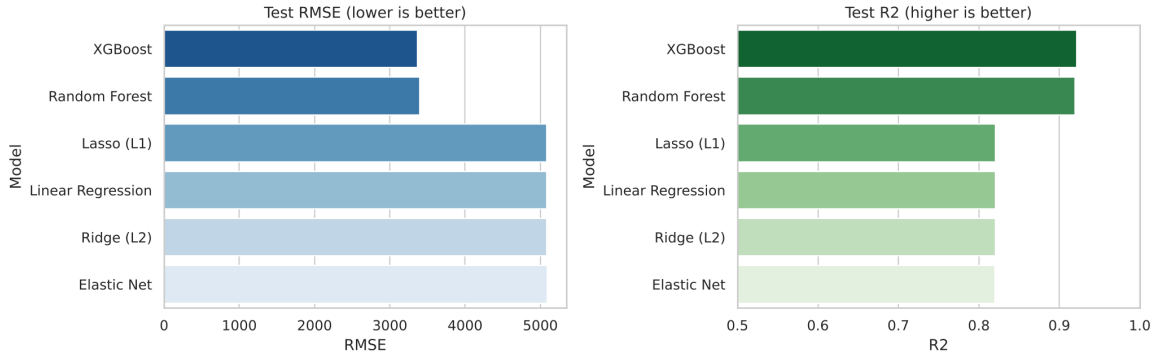


Figure 3. Test RMSE (left) and R-squared (right) by model.

3.1 Interpretation

- **Linear family (Linear, Ridge, Lasso, Elastic Net)** are practically identical (RMSE about $5,086, R2 0.821). With only 8 encoded features and little multicollinearity, regularisation has nothing to fix: the best alpha values were small and Lasso removed no feature.
- **Tree ensembles clearly win.** XGBoost reduces RMSE by 34% and MAE by 41% relative to linear regression (RMSE $3,364 vs $5,086). Random Forest is almost as good (RMSE $3,400); the difference between the two is smaller than the CV standard deviation, so they are statistically close.
- **Why:** cost depends on interactions (smoker x BMI x age). When we hand the linear model two engineered features, its RMSE falls to $3,440 (R2 0.918), almost matching XGBoost - proof that the gain comes from capturing the interaction, not from model complexity as such.

- **Business translation:** XGBoost's MAE of $2,120 means a typical prediction is off by about $2,120 against an average cost of $13,279. The larger RMSE ($3,364) reveals that a few expensive customers are still mis-estimated by much more than that.
- Adjusted R2 (0.919) is very close to R2 (0.922), so the model is not inflated by useless variables.

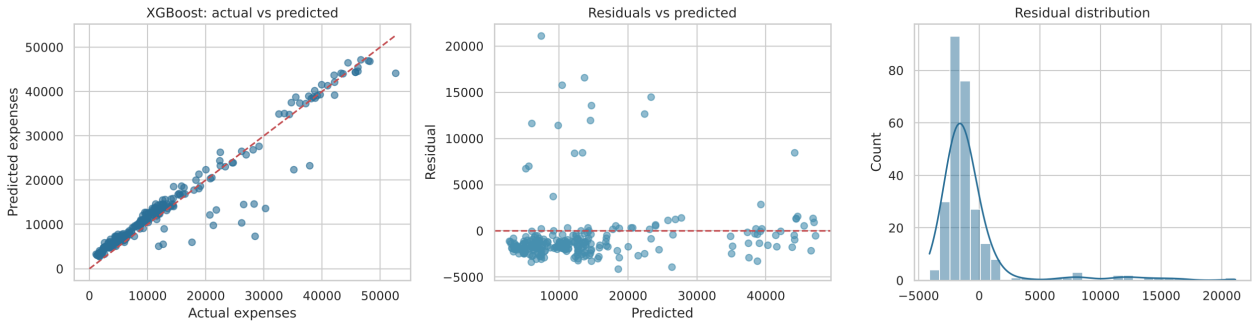


Figure 4. XGBoost diagnostics: predictions hug the diagonal; residuals are centred on zero with a small number of large misses (right tail).

4. Classification Results: High-Risk vs Low-Risk Customers

The test set contains 268 customers, of whom only **27** are high risk, so one missed customer changes recall by 3.7%. For this reason the cross-validated columns (CV Recall, CV F2, computed on 5 stratified folds of the training data) are the more reliable basis for choosing a model.

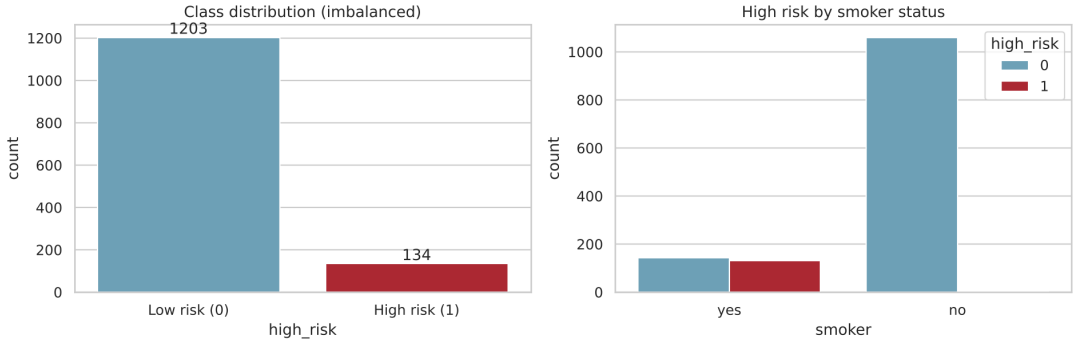


Figure 5. Class imbalance: about 10% of customers are high risk. 48% of smokers are high risk versus 0.3% of non-smokers.

Model	Strategy	Acc.	Prec.	Recall	F1	AUC	CV Recall	CV F2
Logistic Regression	None (baseline)	0.985	0.960	0.889	0.923	0.991	0.831	0.851
	Class weights	0.933	0.615	0.889	0.727	0.990	0.971	0.853
	SMOTE	0.951	0.706	0.889	0.787	0.990	0.971	0.876
	Undersampling	0.918	0.551	1.000	0.711	0.992	0.971	0.823
Decision Tree	None (baseline)	0.981	1.000	0.815	0.898	0.940	0.887	0.900
	Class weights	0.963	0.774	0.889	0.828	0.942	0.953	0.922
	SMOTE	0.985	1.000	0.852	0.920	0.942	0.934	0.923
	Undersampling	0.981	0.923	0.889	0.906	0.982	0.944	0.935
Random Forest	None (baseline)	0.985	1.000	0.852	0.920	0.991	0.888	0.898
	Class weights	**0.981**	**0.923**	**0.889**	**0.906**	**0.993**	**0.953**	**0.938**
	SMOTE	0.981	0.923	0.889	0.906	0.995	0.944	0.933
	Undersampling	0.981	0.923	0.889	0.906	0.992	0.953	0.933
XGBoost	None (baseline)	0.989	1.000	0.889	0.941	0.968	0.916	0.919
	Class weights	0.981	0.923	0.889	0.906	0.955	0.934	0.930
	SMOTE	0.989	1.000	0.889	0.941	0.951	0.934	0.932
	Undersampling	0.970	0.828	0.889	0.857	0.980	0.944	0.923

Table 2. All 16 model / strategy combinations (test-set metrics at threshold 0.5, plus cross-validated recall and F2). Selected model highlighted.

4.1 Effect of the imbalance strategies

- **Accuracy is misleading:** every baseline reaches 98 to 99 % accuracy, yet the naive 'always low risk' rule already scores 89.9%. Accuracy hides the real question - how many risky customers are caught.
- **Cross-validated recall improves with imbalance handling** for every model: Logistic Regression 0.83 to 0.97 (class weights), Decision Tree 0.89 to 0.95, Random Forest 0.89 to 0.95, XGBoost 0.92 to 0.93.
- **The price is precision:** for Logistic Regression, class weights, SMOTE and undersampling cut test precision from 0.96 to between 0.55 and 0.71 - more safe customers get flagged for review. Tree ensembles lose far less precision.
- **Test-set differences between strategies are small** (often one customer) and should not be over-read. In this dataset smoking status separates the classes so strongly that even baseline models perform well; on a dataset with a weaker signal and a 5 % claim rate the benefit of SMOTE and class weights would be much larger.

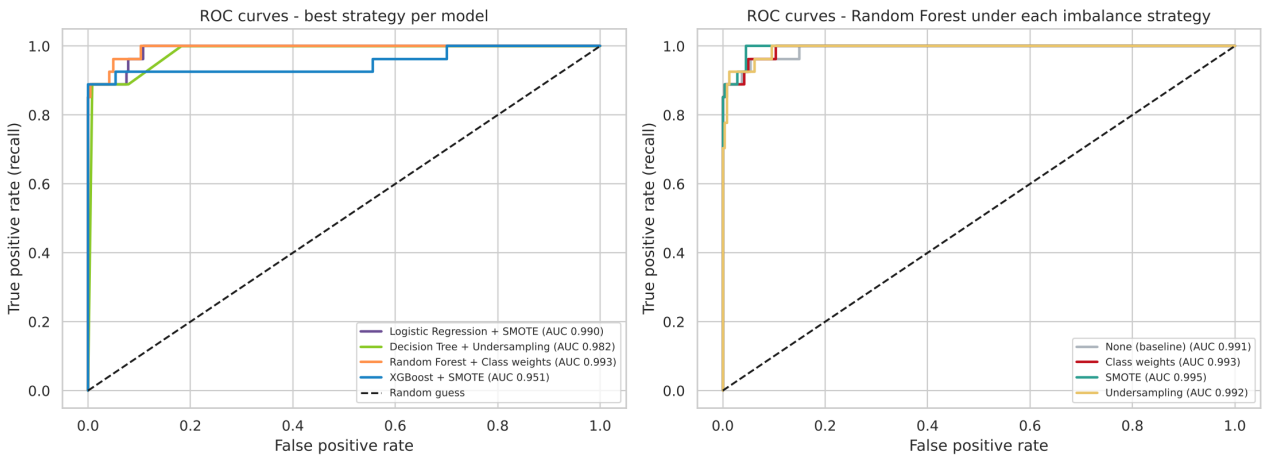


Figure 6. ROC curves: best strategy per model (left) and Random Forest under each imbalance strategy (right). All strong models reach AUC above 0.95.

4.2 Selected classifier

Selecting on cross-validated F2 (recall counts twice as much as precision) picks **Random Forest with class weights** (CV F2 0.938). On the test set it achieves precision **0.923**, recall **0.889**, F1 0.906 and ROC-AUC **0.993**. The confusion matrix shows 24 true positives, 3 false negatives (missed high-risk customers), 2 false positives and 239 true negatives.

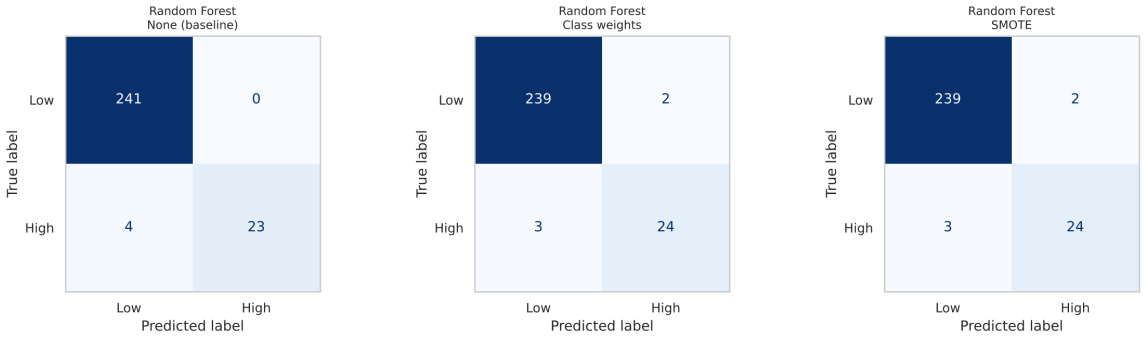


Figure 7. Confusion matrices: Random Forest baseline (left), selected model (centre), SMOTE variant (right).

4.3 Choosing the decision threshold

The 0.5 cut-off is a business choice, not a law. Table 3 shows what happens if the underwriting team lowers it.

Threshold	Precision	Recall	Flagged for review	High-risk missed
0.2	0.650	0.963	40	1
0.3	0.828	0.889	29	3
0.4	0.923	0.889	26	3
0.5	**0.923**	**0.889**	**26**	**3**
0.6	0.960	0.889	25	3

Table 3. Threshold trade-off for the selected model (test set, 27 high-risk customers).

Lowering the threshold to 0.2 catches 96.3% of high-risk customers but flags 40 people for review (precision 65%). If a missed high-risk customer costs much more than a manual review, a threshold of 0.2 to 0.3 is justified; this should be set with the underwriting team using real cost figures.

5. Model Explainability with SHAP

Regulators and customers can demand to know *why* a particular price or decision was produced, so black-box models are not acceptable on their own. SHAP (SHapley Additive exPlanations) uses game theory to compute how much each feature moved an individual prediction up or down relative to the average prediction.

5.1 Regression model (XGBoost)

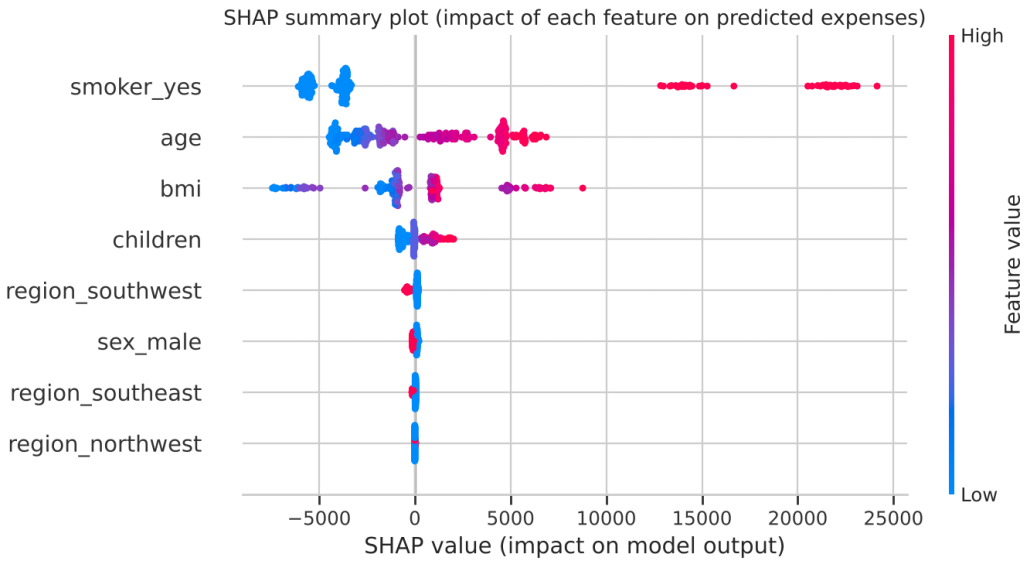


Figure 8. SHAP summary plot. Each dot is a customer; red = high feature value. Right of zero = raises predicted cost.

- **smoker** is the dominant feature: mean absolute impact about $7,407 per prediction. Smokers (red dots) are pushed far to the right.
- **age** ($3,181) and **bmi** ($2,049) follow; older age and higher BMI steadily raise the predicted cost.
- **children** contributes about $595; **sex** ($102) and the **region** dummies (below $188) are nearly irrelevant.

5.2 Force plots for individual customers

High-cost customer: 53-year-old female, BMI 36.9, smoker, 3 children. Predicted $47,183 (actual $46,661). Smoking, BMI and age all push the prediction up from the average of about $13,279.

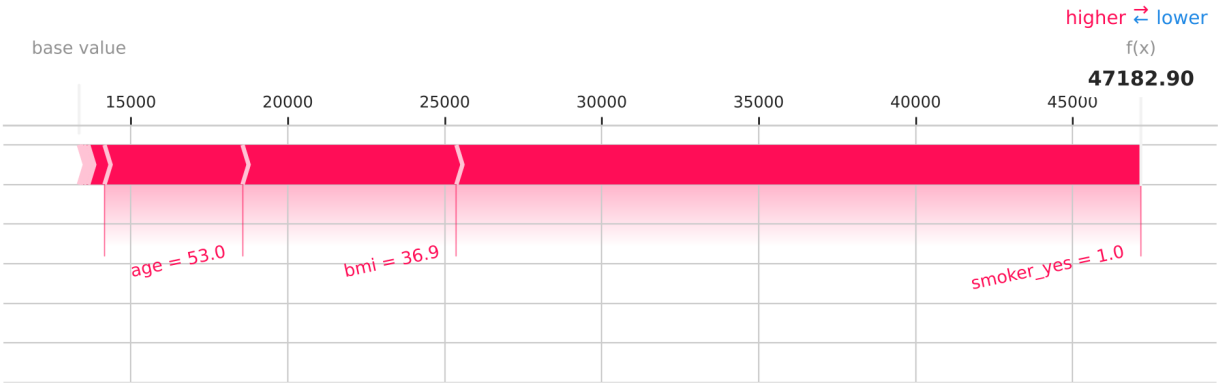


Figure 9. SHAP force plot for the highest-cost customer: red features increase the predicted expense.

Typical low-cost customer: 28-year-old female, BMI 28.9, non-smoker. Predicted $6,286 (actual $4,338); being a young non-smoker pushes the prediction below the average.

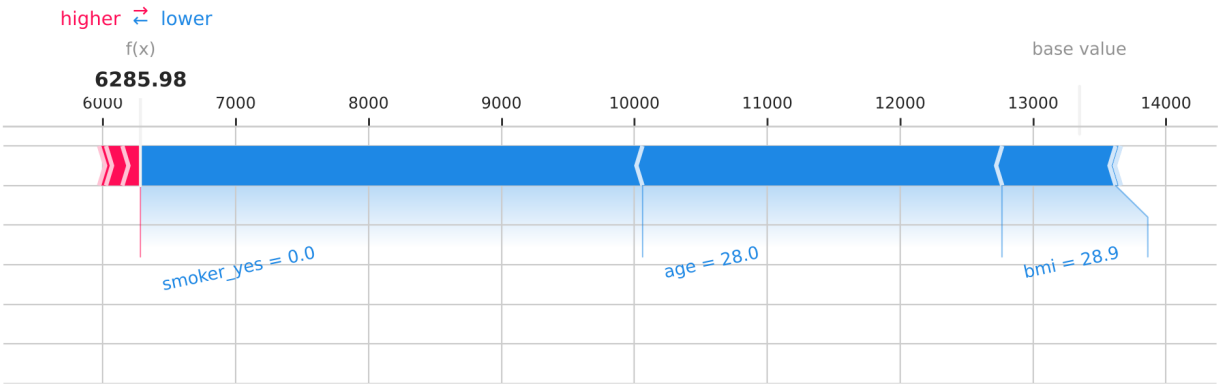


Figure 10. SHAP force plot for a low-cost customer: blue features decrease the predicted expense.

5.3 Classification model (Random Forest)

The classifier relies on the same factors: mean absolute SHAP values are **smoker 0.320**, **bmi 0.160** and **age 0.024**, with sex, children and region below 0.011. Smoking and obesity together are what make a customer high risk.

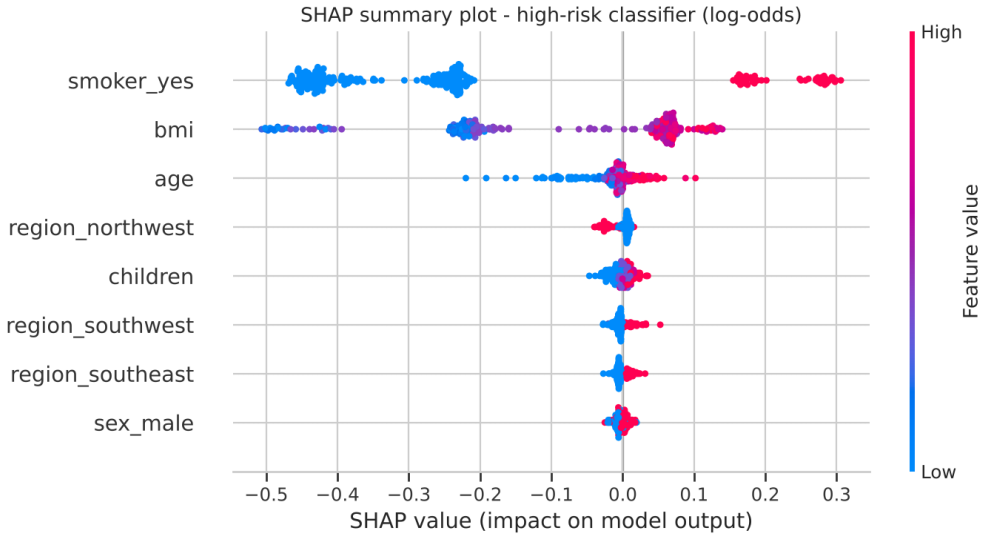


Figure 11. SHAP summary for the high-risk classifier.

The force plot below explains one flagged customer: a 47-year-old male smoker with BMI 36.1 receives a high-risk probability of 0.96 (actual label: high risk). Smoking and BMI contribute almost all of the increase over the base value.

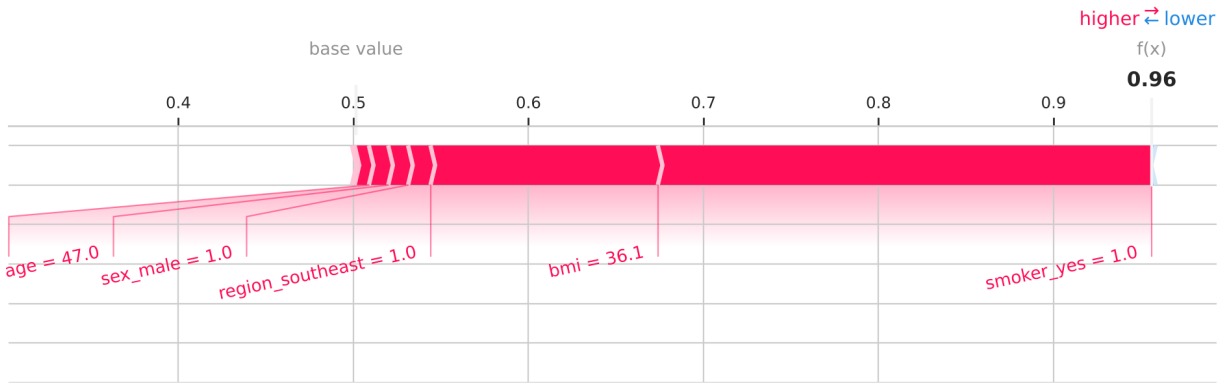


Figure 12. SHAP force plot for one customer flagged as high risk (output = probability).

5.4 Why this supports compliance

Every prediction can be accompanied by 'reason codes' taken directly from its SHAP values (illustrative format: 'smoker status +X dollars; BMI +Y dollars; age +Z dollars'). Because sex and region contribute very little, the model can be retrained without them, which reduces fairness and regulatory risk at almost no cost in accuracy.

6. Model Selection and Deployment Recommendation

6.1 Recommended models

Task	Recommended model	Key result (test set)	Reason
Premium / cost prediction	**XGBoost Regressor**	RMSE $3,364, MAE $2,120, R2 0.922	Lowest error, stable under cross-validation, explainable with SHAP
High-risk triage	**Random Forest (class weights)**	Recall 0.89, precision 0.92, AUC 0.993	Best cross-validated F2; keeps precision high while catching most risky customers
Challenger / fallback	**Linear regression + interaction features**	RMSE $3,440, R2 0.918	Almost as accurate and fully transparent; useful as a benchmark and a regulator-friendly fallback

6.2 Proposed deployment

- **Two-stage workflow:** the classifier screens every application. Low-probability customers are approved automatically; flagged customers go to an underwriter together with the SHAP explanation. The regressor then supports pricing.
- **Human in the loop:** keep underwriters responsible for final decisions on flagged cases and on any customer who disputes a price.
- **Threshold governance:** set the classification threshold with the business using real costs of a missed high-risk customer versus a manual review (Table 3).
- **Monitoring:** track error (MAE/RMSE), recall and the input distribution monthly; retrain at least yearly or when drift is detected (for example a change in smoking rates or medical costs).
- **Documentation:** store the SHAP reason codes, model version and training data snapshot for every decision to support audits.

6.3 Limitations and risks

- **Small data:** only 1,337 customers and 27 high-risk customers in the test set, so test-set metrics carry a wide margin of error (one customer = 3.7% of recall).
- **Label definition:** "high risk" was defined by an expense percentile because no claim-event label exists. A real project would use actual claim outcomes and loss ratios.
- **Expenses are not premiums:** the target is a cost proxy; real premiums also include expenses, profit margin and regulation.
- **Limited features:** no medical history, driving or telematics data; smoking status is self-reported and can be misreported.
- **Fairness and law:** using sex and region in pricing is restricted in many jurisdictions. Their SHAP contribution is tiny, so they should be tested for removal and the model audited for disparate impact.

- **Extrapolation:** tree models cannot predict outside the range of the training data (for example costs far above the observed maximum).

6.4 Conclusion

Smoking status, age and BMI explain most of the cost and risk in this portfolio. Non-linear ensembles capture their interactions and cut prediction error by about one third compared with a plain linear model. Combined with SHAP explanations, class-imbalance handling and a recall-oriented threshold, the proposed solution is accurate, explainable and practical to deploy under human supervision.

Appendix: Reproducibility

- **Software:** Python 3, pandas, scikit-learn, imbalanced-learn (SMOTE, undersampling), XGBoost, SHAP, matplotlib, seaborn.
- **Random seed:** 42 for all splits, cross-validation and models.
- **Notebooks:** 01_Regression_Insurance_Expenses.ipynb and 02_Classification_Insurance_Risk.ipynb. Run all cells in order with med-insurance.csv in the same folder.
- **All numbers in this report** were produced by executing those notebooks; none were typed in by hand.

Additional figures

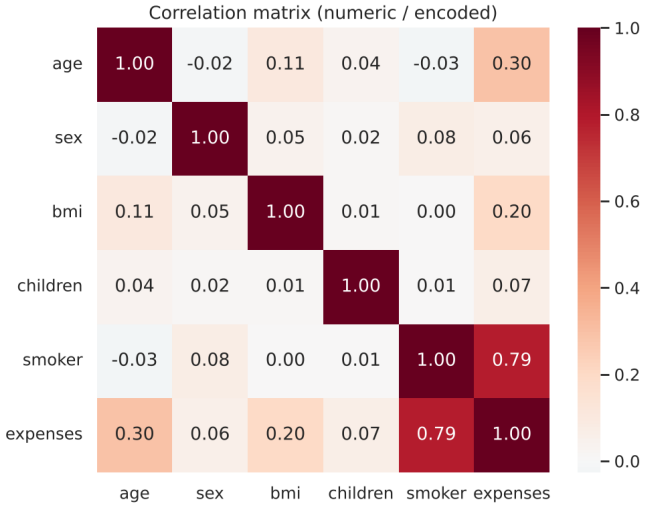


Figure A1. Correlation matrix: smoker (0.79) is the dominant correlate of expenses, then age and BMI.

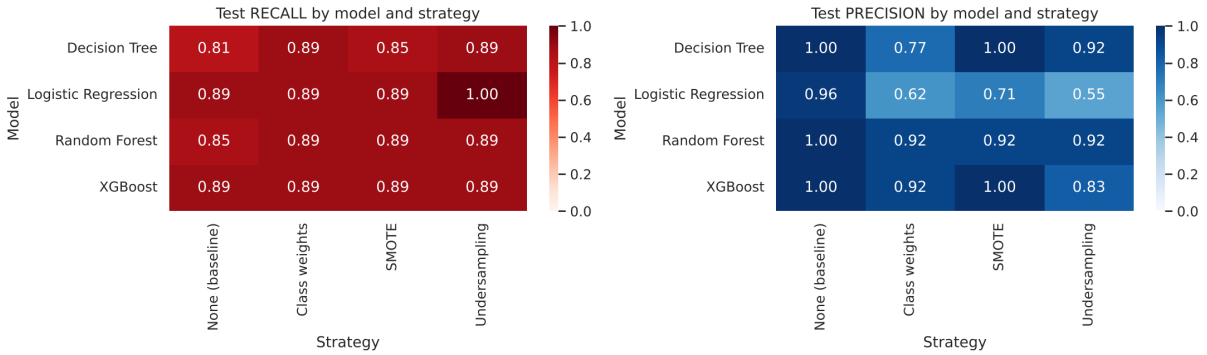


Figure A2. Test recall (left) and precision (right) for each model and imbalance strategy.

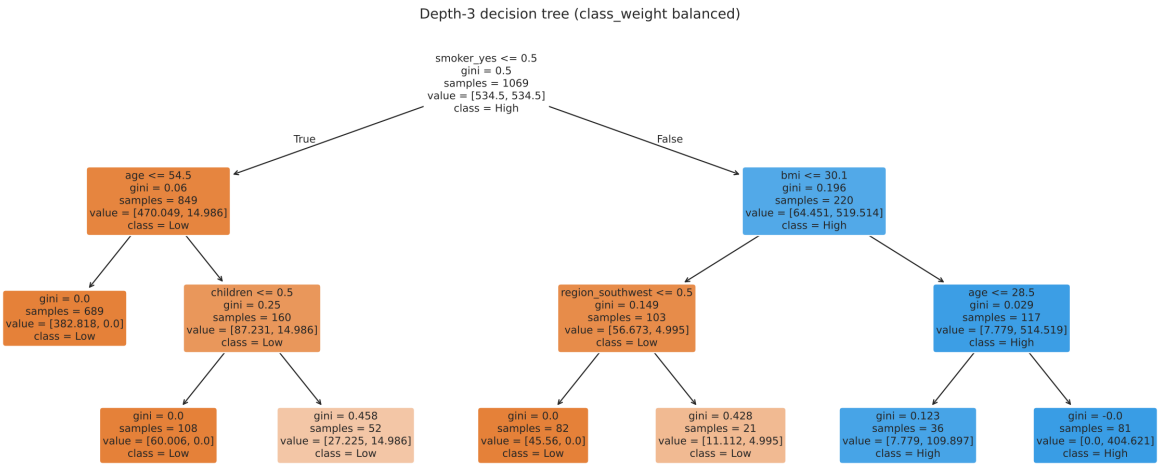


Figure A3. A depth-3 decision tree: easy to read, but an unrestricted tree overfits (see notebook 2, section 8).